

tmux documentation

basic-digital tmux — QWEN.md

Qwen Code CLI agent view of this project's governance. Base agent rules live at `constitution/QWEN.md` and the `constitution/Constitution.md` it references. **READ THOSE FIRST.** This file extends them with project-specific rules; it never weakens them. Canonical project authority: [Constitution.md](#) — Project Articles §101–§109. For the full agent rule set see [AGENTS.md](#), which this file mirrors.

Fresh-conversation workflow: read `CONTINUATION.md` first (§0/§3/§8 mandatory), then `Issues.md` for `OPEN/PARTIAL/BLOCKED/RUNNING`.

MANDATORY ANTI-BLUFF END-USER-QUALITY COVENANT

Forensic anchor — verbatim user mandate (2026-04-28, reasserted 2026-05-21):

"We had been in position that all tests do execute with success and all Challenges as well, but in reality the most of the features does not work and can't be used! This **MUST NOT** be the case and execution of tests and Challenges **MUST** guarantee the quality, the completion and full usability by end users of the product!"

Operative rule. The bar for shipping is **not** "tests pass" but **"end users can actually use the feature."** Every `PASS` — test OR HelixQA Challenge — **MUST** carry positive captured runtime evidence that the feature works for the end user. Metadata-only `PASS`, configuration-only `PASS`, "absence-of-error" `PASS`, and `grep-without-runtime` `PASS` are all critical defects regardless of how green the summary line looks. Tests and Challenges are bound **EQUALLY**.

This covenant is restated verbatim in every governance file at the consumer layer (`CLAUDE.md`, `AGENTS.md`, `QWEN.md`) so any tool that does not expand `@imports` still reads it. The same verbatim block lives upstream at `constitution/Constitution.md` §11.4, `constitution/CLAUDE.md`, and `constitution/AGENTS.md`. Removing or weakening this block is a §11.4-class release blocker.

Canonical authority: `constitution/Constitution.md` §11.4 and its sub-anchors §11.4.1 through §11.4.78. Project anchor: [Constitution.md](#) §101.

Non-compliance is a release blocker regardless of context.

Project overview

Verified hardened tmux 3.6a build with jemalloc, OOM protection, and per-session isolation via the `tmx` wrapper. Native dual-OS: each `tmx new -s NAME` spawns its own tmux server with OS-native isolation — `cgroup-v2` scope on Linux, `POSIX rlimit` wrapper on macOS. Session shell is the operator's host shell with full `PATH`. No VM.

Critical base rules restated (for agents that don't follow @imports)

- **No bluffing.** Every PASS — test OR HelixQA Challenge — carries positive runtime evidence (file *content*, not existence). Universal §11.4 / Project §101.
- **FAIL-bluffs forbidden.** Fix script-bug failures at the source layer, never at call sites. Universal §11.4.1.
- **No-guessing language.** likely, probably, maybe, seems, appears are forbidden when reporting causes — prove with forensic evidence or mark UNCONFIRMED: / PENDING_FORENSICS:. Universal §11.4.6.
- **Operator-path coverage.** Gate tests use `tmx new -s X`. Project §102.
- **Four-layer coverage.** Every defect lands pre-build gate + runtime test + HelixQA Challenge + paired mutation. Project §103.
- **Use the commit wrapper.** `bash commit_all.sh "<msg>"` only — no direct `git push` on the main repo.
- **Never force-push** without explicit per-operation authorization.
- **CONTINUATION.md** kept in sync in every non-trivial commit. Universal §12.10.
- **60 % RAM cap** enforced per-session via `TMX_MEM`. Project §106.

Commands (exact)

Step	Command
Install build deps	<code>bash scripts/install_deps.sh</code>
Full pipeline	<code>bash scripts/setup.sh</code>
Verification gate	<code>bash scripts/verify.sh</code>
Run all tests	<code>bash scripts/tests/run_all.sh</code>
Inheritance gate	<code>bash scripts/tests/ test_constitution_inheritance.sh</code>
Meta-test (paired mutation)	<code>bash scripts/tests/ meta_test_false_positive_proof.sh</code>
End-to-end automation	<code>bash scripts/test_e2e.sh</code>
Commit + push (github+gitlab)	<code>bash commit_all.sh "message"</code>
Per-session wrapper	<code>tmx {new attach ls kill}</code>

Project-specific agent rules

Forbidden in this project

- Editing `tmux/`, `constitution/`, or generated files (`scripts/tmx`, `tmux/build*/`).
- Direct `git push` on the main repo — use `commit_all.sh`.
- CI/CD pipeline files; HTTPS Git remotes (SSH only).
- Advancing the `tmux/` submodule off tag 3.6a without an explicit documented decision (Project §107).

Project-specific workflow

- **Test-interrupt-on-discovery** (§103): a defect found mid-cycle STOPS the cycle — fix root cause, land four layers, rebuild, repeat.
- **Topology dispatch** (§104): tests detect host topology and SKIP-with-reason; never silently degrade.
- **Destructive tests** (12/13/14): require `TMX_TEST_DESTRUCTIVE=1`.

Project overrides of universal rules

None.